A completely specified Diophantine consistency equation

Fifteen strictly positive integer unknowns; all fixed constants defined on this page.

The unknowns are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

Use the following polynomial abbreviations (no extra unknowns):

$$\begin{gathered} q = \beta J + 1, \quad C = q - F - \alpha, \quad X = wq, \\ Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \quad c = kY + \eta, \\ a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\ D = X + ac + \gamma H, \\ R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\ V = c(Tf - 1) - Rf^2. \end{gathered}$$

The equation is

$$\boxed{N_0 N_1 N_a N_k N_t N_s - 1 = 0,}$$

where

$$\begin{aligned} N_0 &= \tau^2 - (XY^2)(XY^2 + 1)k^2, \\ N_1 &= D^2 - \Delta c^2, \\ N_a &= \Delta(f^2 - 1)(V^2 - y^2) + y^2, \\ N_k &= k - hE - R, \\ N_t &= (K + w)C + (q - F) - z(q - 1), \\ N_s &= f^2 - \Delta(ic^2)^2. \end{aligned}$$

Fixed coefficients. All sums below have the displayed fixed finite limits. Division and floor are used only to define constants; the equation above is polynomial. Put

$$\begin{gathered} \rho = 2^{89}, \quad B = \rho^{4831}, \quad \beta = B - 1, \\ \lambda = 2^{29}, \quad \theta = 2^{42}, \quad Q = 2^{40}, \end{gathered}$$

$$G_m(x) = \sum_{j=0}^{m-1} x^j,$$

$$\mu = 510 + 2^{17} + 2^{26} + 2^{28} + \lambda(2^{44} - 1).$$

Let

$$\begin{aligned} \mathcal{S} = {} & \{0, \dots, 456\} \cup \{495, \dots, 504\} \cup \{507, 508\} \\ & \cup \{517, \dots, 537\} \cup \{539, \dots, 549\} \\ & \cup \{567, \dots, 583\} \cup \{1203, 1807\}. \end{aligned}$$

Then

$$\begin{aligned} M_C &= B - 1 - \sum_{e \in \mathcal{S} \setminus \{1\}} \rho^e - 2\rho^{507}, \\ M_F &= B - 1 + 4 + \mu(\rho^{2409} + \rho^{3015}) \\ &\quad + \rho^{528} G_{10}(\rho) + \rho^{585} G_{11}(\rho) + \rho^{1203} + \rho^{1807} \\ &\quad + (2^{87} - 1)\rho^{4823} G_4(\rho) + (2^{38} - 1)\rho^{4827}, \\ K &= \rho^{33} + \rho^{68} + \rho^{664} + \rho^{1268} + \rho^{4877} \\ &\quad + (\rho^{2409} + \rho^{3015})\Phi(1/\rho), \end{aligned}$$

where the rational intermediate expression is

$$\begin{aligned} \Phi(x) = {} & \sum_{i=0}^{456} c_i x^{r_i} \\ & + \lambda\big[(x^{539} + 2^{21}x^{517})G_{11}(2x) \\ & \quad + (2^{11}x^{528} + 2^{32}x^{495})G_{10}(2x) \\ & \quad + \theta(x^{539} + x^{1203} + x^{1807})\big] \\ & + \big[(2^9 + \lambda(2 + 2^{33}))x^{567} \\ & \quad + (2^{18} + \lambda(2^{22} + 2^{33}))x^{575}\big]G_8(2x) \\ & + (2^{27} + \lambda(2^{11} + 2^{32}))x^{583}. \end{aligned}$$

The resulting β, K, M_C, M_F are fixed positive integers.

The constants r_i, c_i. First define the following 457 fixed quadruples $t_i = (w_i, n_i, e_i, s_i)$. From the integer $\mathcal{A}$ printed below, for $0 \le j < 430$ put

$$\begin{gathered} b_j = \left\lfloor \mathcal{A}/1024^j \right\rfloor \bmod 1024, \quad u_j = b_j \bmod 256, \\ v_j = \lfloor b_j/256 \rfloor \bmod 2, \quad d_j = \lfloor b_j/512 \rfloor. \end{gathered}$$

Here $a \bmod b = a - b\lfloor a/b \rfloor$. Set

$$t_{2+j} = \big((1 - d_j)(512 + u_j),\ 512 + j,\ d_j(256 + u_j),\ v_j\big).$$

The remaining quadruples are

index	range	t_i
0		$(2, 3, 3, 1)$
1		$(5, 0, 0, 0)$
$432 + j$	$0 \le j < 2$	$(0, 942 + j, 0, 942 + j)$
$434 + j$	$0 \le j < 2$	$(0, j, 0, j)$
$436 + j$	$0 \le j < 4$	$(0, (j \bmod 2) + 942\lfloor j/2 \rfloor, 0, 6)$
440		$(2, 2, 2, 0)$
441		$(3, 4, 4, 698)$
442		$(4, 4, 4, 0)$
$443 + j$	$0 \le j < 3$	$(1, 6, 1, 2 + j)$
$446 + 3j + k$	$0 \le j, k < 3$	$(j^2, 5, k, 5)$
455		$(256, 0, 0, 512)$
456		$(0, 0, 512, 512)$

Define, for these fixed indices,

$$\begin{aligned} \kappa_i &= 2^{30}w_i + 2^{20}n_i + 2^{10}e_i + s_i, \\ \ell_i &= 2w_i + 2^{11}n_i + 2^{22}e_i + 2^{32}s_i, \\ r_0 &= 0, \quad r_1 = 1, \\ r_i &= 2 + \sum_{j=2}^{456} \left\lfloor \frac{Q + \kappa_i - \kappa_j - 1}{Q} \right\rfloor \quad (2 \le i \le 456), \\ \delta_{ij} &= \left\lfloor \frac{Q - (i - j)^2}{Q} \right\rfloor, \\ O &= 1 + 255\,2^9 + 255\,2^{18} + 2^{27} + \lambda, \\ c_i &= O + \lambda\ell_i - 255\,2^9\delta_{i,455} - 255\,2^{18}\delta_{i,456} \\ &\qquad - 2^{27}(\delta_{i,455} + \delta_{i,456}). \end{aligned}$$

All subscripts in these definitions are bound constant indices, not additional Diophantine unknowns.

The integer $\mathcal{A}$. Concatenate the following lines in order:

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56118335615737617036884126988661458925244389493947058121715971
01282916856878425443037240736299803427824619599002153617345708
28900551636239650126542082909466175026205487572409217714429639
74793862932436776448290221360877005574415446786244745611959287
59373112491605173997640015648449178599058708778363556006068759
35678639237640904034915105614223460060014840095836650944470526
30060761226207037490030688247103632272616121659441631458841739
88053038481633961651885200391264203970434788452535325946647905
26112698210284600353636434795855158687861272006223054621338052
08060289634159601137749386030632145140305682244017830050480315
09861375430770737373574647116320836883093705981192684117359643 8
99567325570145003948256119536072783258394320179887629332814913
83742105792556854838396752391719155979103190193016087490398924
39017272426043275053833387572790002456817641447316642739994186
97003064581790026300137955120486268219145788062206957924754981
34155288871384611877132683517463256922006640654344797535262481
90435827191666264404447576661943567063783141522216477014676593
40310593002091093201099250164766043944383508572504290831392543
80006501642309702444745828581986468804777321875277055695293159
15172377582008682228504985026248220701674081629382752859968808
92794082915247799854983441973097851601253822624248713 3
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Absence of positive integer solutions is equivalent to Con(ZFC) by the inherited native/Pell argument and the checked P69–P67 reduction. The full native calibration and its internal ZFC formalization remain open Lean obligations. This page specifies the equation, not its proof.